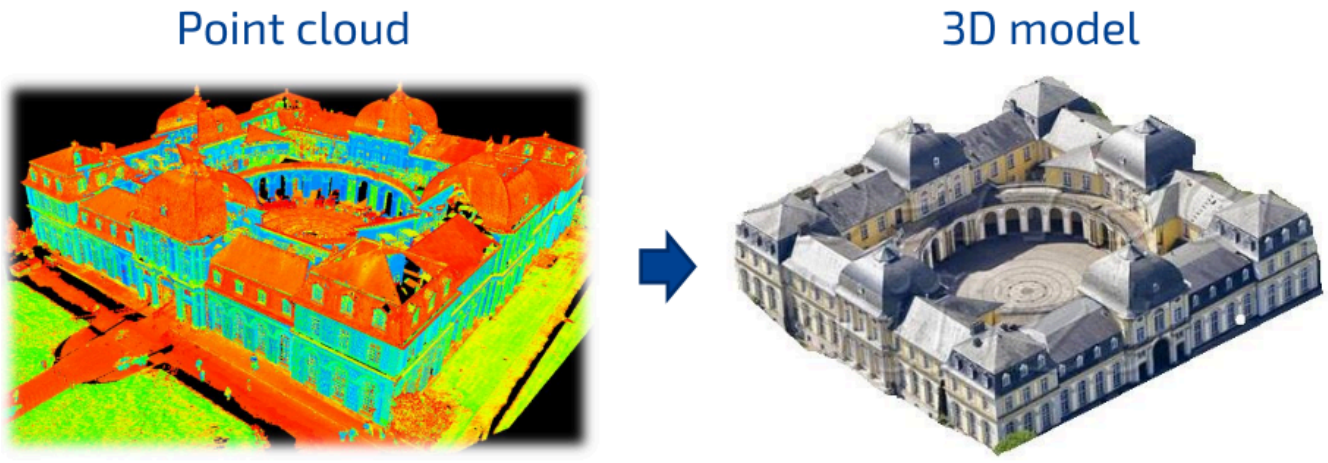
L2: Coordinate Transformations applied to Geodetic Sensors

Combination of measurements of stations into single coordinate system
- Local transformation between sensors (registration)

Terrestrial laser scanner (TLS)

- Need for Coordinate Transformation

Step 1: TLS Registration

- Merging to one single point cloud <=> Coordinate transformations
- **Registration** to local coordinate system x, y, z from diffent local CS
  - arbritary local CS
  - Mathematical approximation
- 

Step 2: Positioning on Earth

- Transformation to global, earth centered CS
- E.g. ITRF or WGS84
- **Geo-referencing**

Step 3: Given Official Coordinates

- Official Coordinates split in position (2D) and height (1D)
- Position is (UTM-)projection due to curved earth
- **Projections**

Sensor CS Realisations

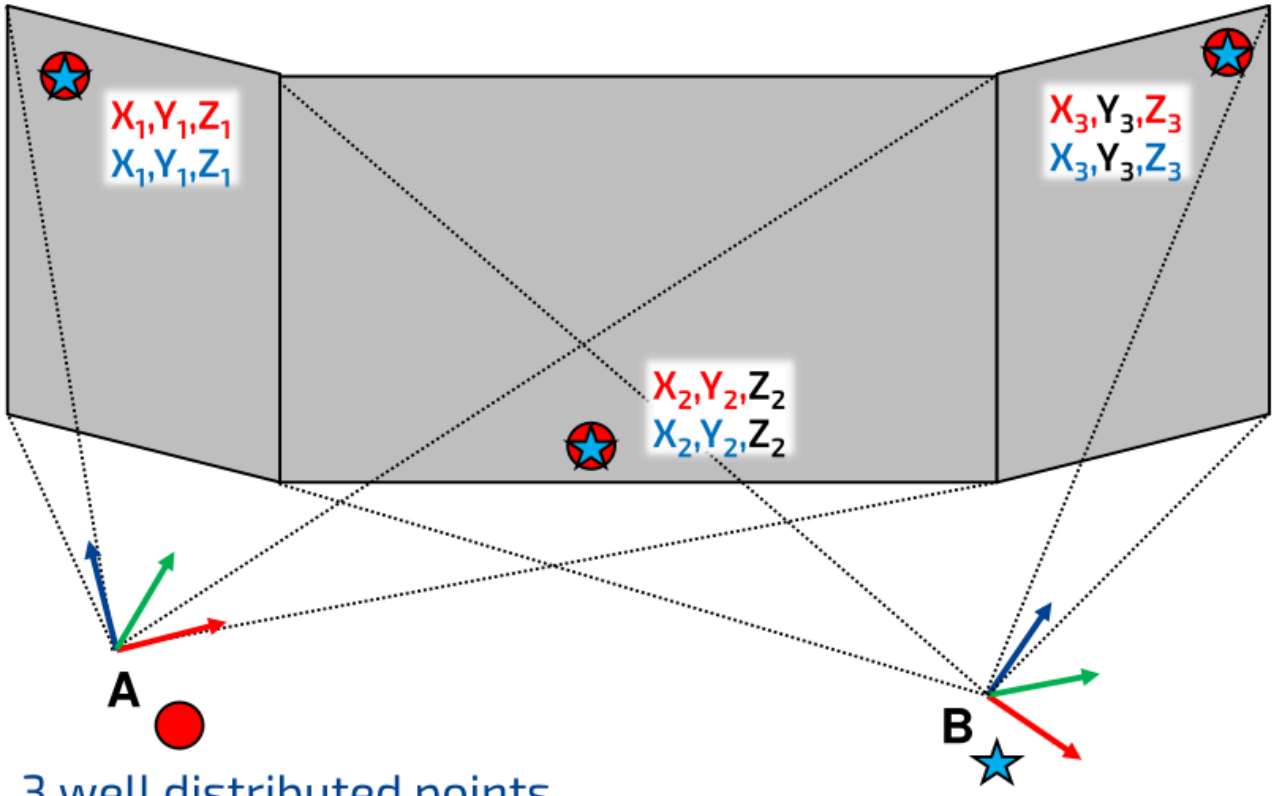
- GNSS Receivers
  - Observables: x, y, z
  - CS: Geocentric, ITRF2014/WGS84
- Camera (Photogrammetry, Structure-from-Motion)
  - Observables: x,y
  - CS: local, image coordinate system
  - Coordinate transformation for 3D reconstruction
  - (Bundle adjustment)
- Total stations
  - Observables: θ, φ, d
  - CS: local, intersection of vertical, horizontal and collimation axis
  - $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d \cdot \sin\theta \cdot \sin\varphi \\ d \cdot \sin\theta \cdot \cos\varphi \\ d \cdot \cos\theta \end{bmatrix}$
- 3D Terrestrial Laser Scanner
  - Observables: $\theta, \varphi, d \rightarrow x, y, z$
  - CS: local, intersection of vertical, horizontal and collimation axis

Registration

Similarity Transform in 3D:

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \mu \boldsymbol{R}_3(\varepsilon_z)\boldsymbol{R}_2(\varepsilon_y)\boldsymbol{R}_1(\varepsilon_x) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

- **7 DoF/ 7 parameters**
  - Scale Change (1DoF)
  - Rotation (3 DoF)
  - Translation (3DoF)
- Determination of these 7 parameters
- 7 or more **observations**(coordinates) in each system
  - 7 = analytical solution
  - > 7 = adjustment
- 3 Well distributed points (otherwise ill posed problem)
- known correspondences



- 3 well distributed points
- 7 or more observations (coordinates) in each system

1. **Artificial targets** (e.g. prism, target, sphere)
   - aimed (e.g. total station) or automatically detected (TLS)
2. **Well- defined keypoints** (e.g. corners)
   - aimed (e.g. total station) or automatically detected (TLS)
3. **Point Clouds**
   - Most often solved by ICP
4. **Geometrical features**
   - Varying in all 3 dimensions
   - modified transformation equation
5. **Alternative**:

- Directly observing transformation parameters with additional harware/sensors
  - Most often used for partial solution (some parameters) - reducing DoF
  - **Reducing number of parameters** increases efficiency and accuracy of result

Reducing DoF

- Time-of-Flight (ToF) for Scale μ
  - EDM (laser): Distance given in meter -> Scale fixed
  - Propagation delay of current atmosphere needs to be known

- Inclinometer for Rotations $\varepsilon_x, \varepsilon_y$
  - If reference system levelled, $\varepsilon_x, \varepsilon_y$ equal inclinations
  - Inclinometer measures: deviations from horizontal plane
  - 3D Sensors often contain compensator
  - Compensator = Inclinometer + unit correcting θ, φ
  - $\varepsilon_x = \varepsilon_y = 0$ if using only sensors with compensator

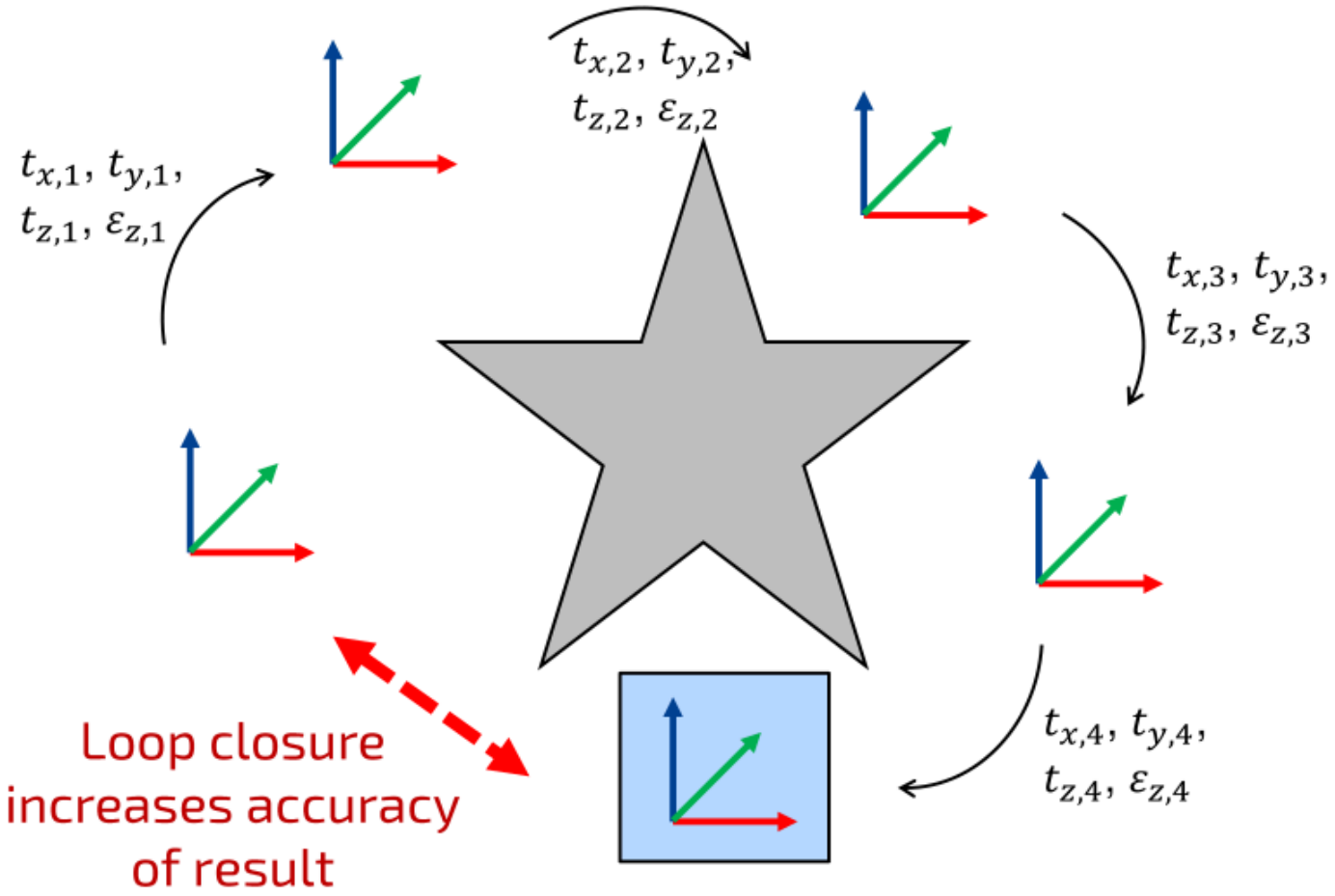
From 7 to 6 or 4 DoF

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda \boldsymbol{R}_3(\varepsilon_z)\boldsymbol{R}_2(\varepsilon_y)\boldsymbol{R}_1(\varepsilon_x) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Most times $\boxed{\lambda = 1}$; $\boxed{\varepsilon_y = 0;\ \varepsilon_x = 0}$ **Often at static data acquisition**

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \boldsymbol{R}_3(\varepsilon_z) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Composition of registrations



Summary

- Using identical artificial points (targets)
- Using measured objects (keypoints, point cloud, geometric features)
- Using additional hardware
- Combination

Geo-Referencing

- Different realisations of global coordinate systems
- Center: Center of certain ellipsoid
- X-axis: Greenwich
- Z-axis: Completion of RHS CS

-> Similarity transformation without Scale -> 6 parameters

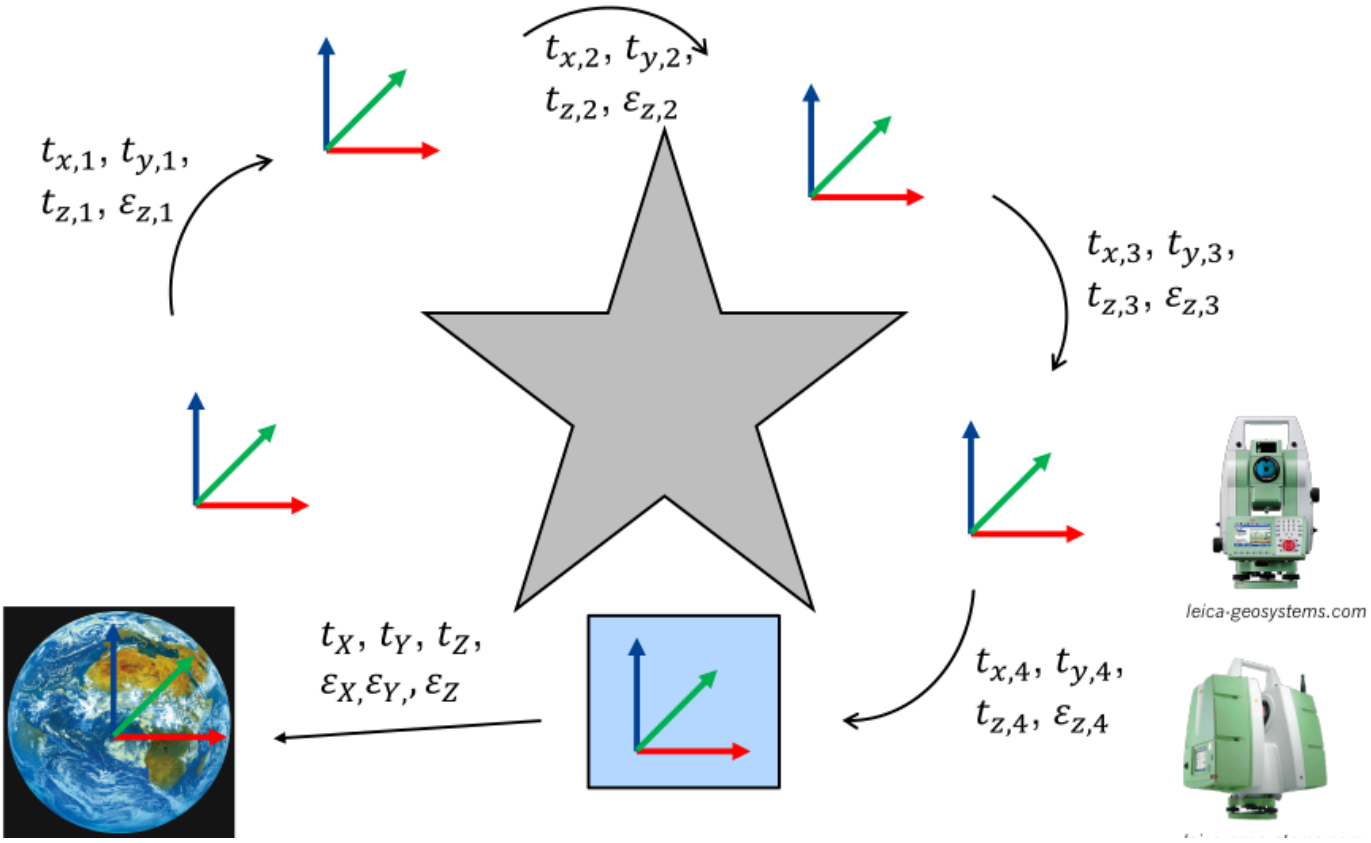
Determination of parameters

- Task for positioning on earth: Transformation to global, earth centered CS
- Determination of paramters:

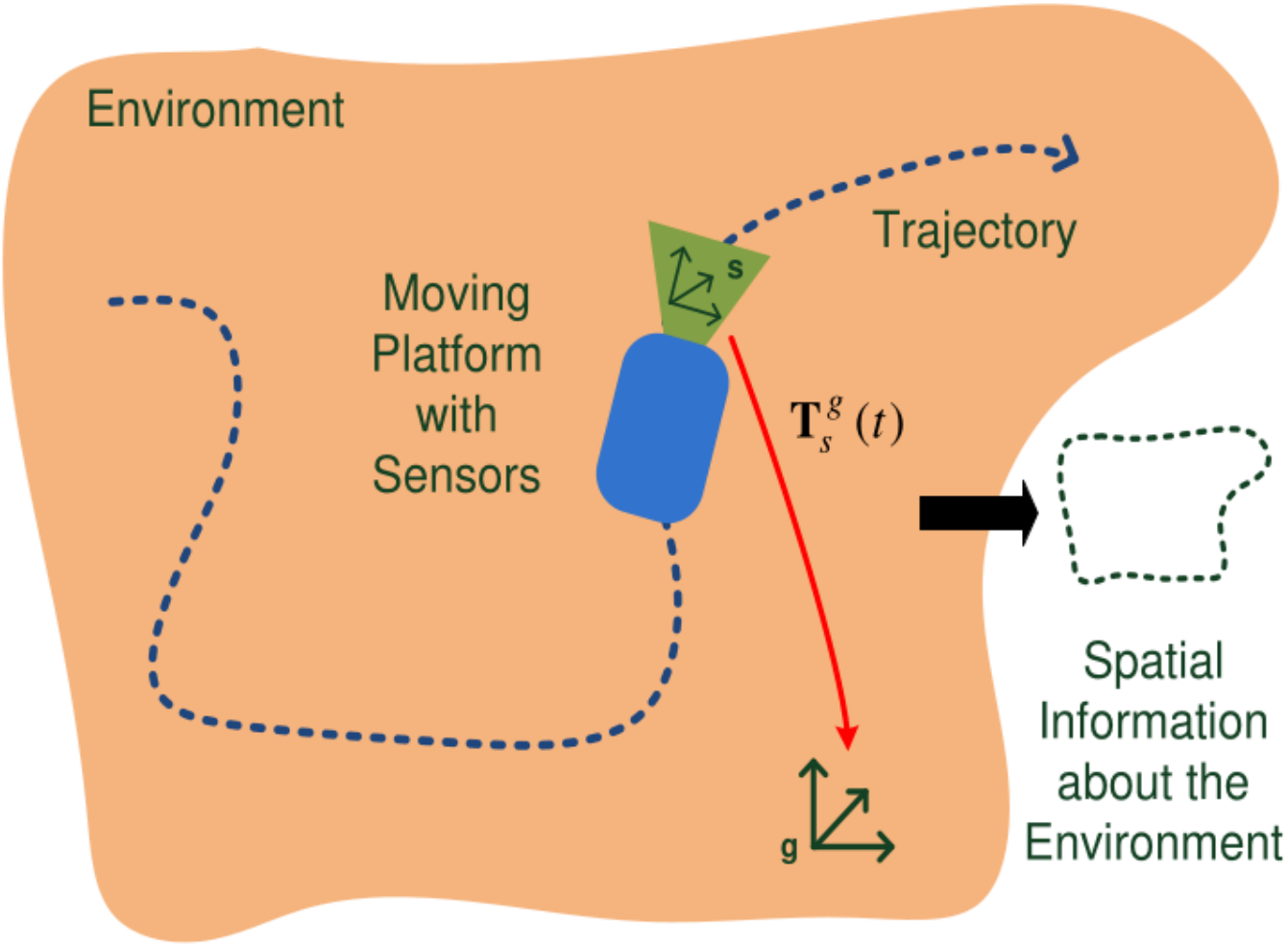
1. Using identical, artificial points, **that are already given in global CS**

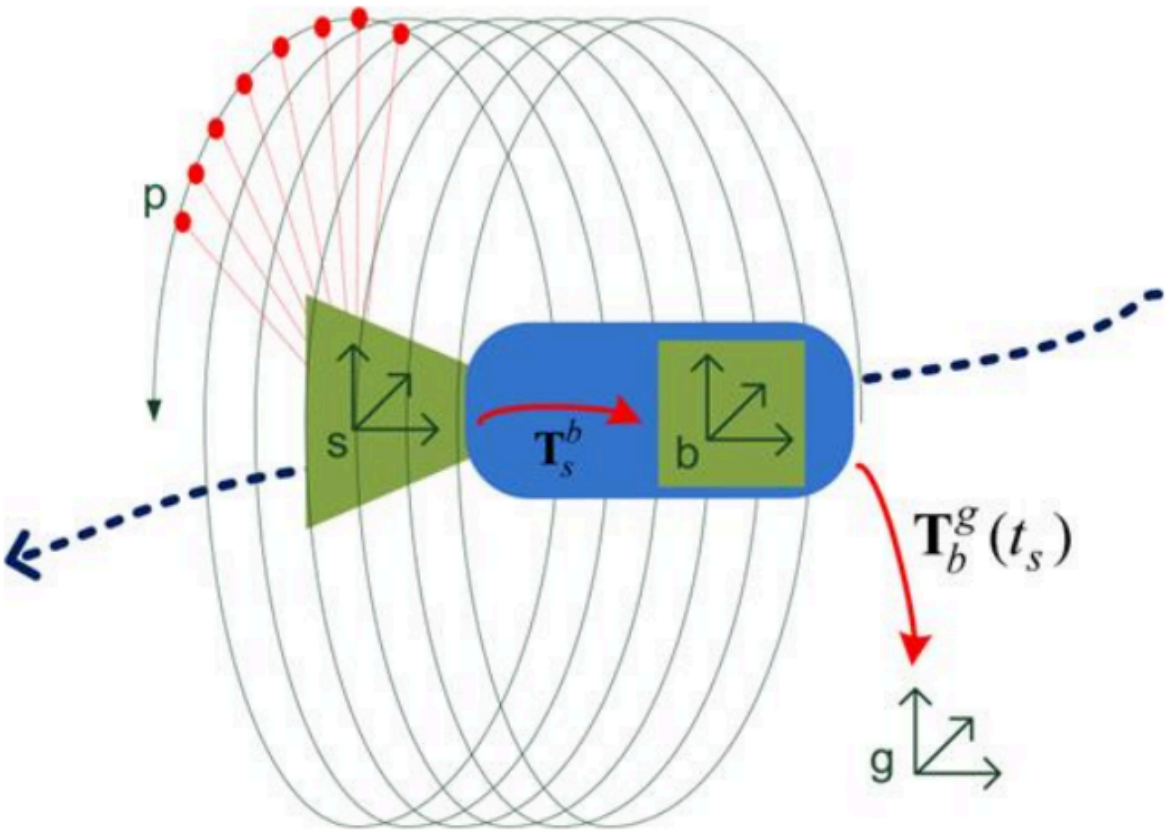
2. Using additional hardware **that acquires data in global CS**
    - GNSS for $t_x, t_y, t_z (most times)
    - Two GNSS for ε_z
    - Inlinometers for $\varepsilon_x, \varepsilon_y$
    - Compass for ε_z
    - Static or kinematic

Static Geo-Referencing



Mobile Multi-Sensor Systems





- **Geo-referencing by:** $\boldsymbol{p}^g = \boldsymbol{T}_b^g(t_s)\boldsymbol{T}_s^b\boldsymbol{p}^s$